PHY102 Electricity and Magnetism Jan-April 2026: Assignment 3

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To be discussed in the Tutorial on Feb 4

1. A point charge is located at the center of a cube of edge length d. What is the value of $\int \vec{E}.d\vec{a}$ over all the different faces of the cube. Now the charge is moved to a one of the corners. How does the flux through each face change? Hint: Using symmetry arguments may be useful.

2. Consider the hydrogen atom where electron charge distribution around the nucleus is given by $\rho(r) = ce^{-2r/a_0}$. a_0(Bohr radius) $= 0.5 \times 10^{-8}$cm and c is a constant.
   - **(a)** Calculate c so that the total charge is the electron charge e.
   - **(b)** Calculate the electric field as a function of distance.
   - **(c)** How much charge is contained in the sphere of radius a_0. Express this as a fraction of e.

3. Consider the gradient of a two dimensional scalar field $f(x, y)$ defined as $\nabla f(x, y) = \frac{\partial f(x,y)}{\partial x}\hat{x} + \frac{\partial f(x,y)}{\partial y}\hat{y}$ By considering a rotated coordinate system which is rotated by an angle θ about the z axis, show that $\nabla f(x, y)$ is indeed a factor. Hint: A vector transforms in a particular way under rotation.

4. Consider a vector field $\vec{F}(x, y) = k(y\hat{x} + x\hat{y})$.
   - **(a)** Calculate the divergence of this vector field.
   - **(b)** Can you associate a potential with this field *i.e.* can you show that the line integral of this field is independent of path?
   - **(c)** Compute the potential function $\phi(x, y)$ such that $\vec{F}(x, y) = -\nabla\phi(x, y)$

5. Consider two spheres of radii R_1 and $R_2 > R_1$ with surface charge density σ on the inner sphere and $-\sigma$ on the outer sphere. Find the electric field in entire space.

6. A sphere with surface charge density σ and radius R expands to a radius $2R$. This process brings what kind of change in:
   - **(a)** Surface charge density.
   - **(b)** Electric field in different regions.
   - **(c)** How much work is done in this process? How does the total energy associated with the electric field change in this process?

7. Does the function $f(x, y) = x^2 + y^2$ satisfy two dimensional Laplace equation? How about $g(x, y) = x^2 - y^2$. Sketch the functions $f(x, y)$ and $g(x, y)$. Calculate the gradient of the functions $f(x, y)$ and $g(x, y)$. Sketch the gradient as arrows. Pay attention to the points $(0, 1), (1, 0), (0 - 1)(-1, 0)$.

8. Consider a ring of radius a with uniform charge density λ with its center at the origin and lying in the $x - y$ plane. Calculate the electric field due to this ring along the z axis. For the region $r >> a$ write the field in such a way that it is an expansion in the powers of a/r. What happens if you keep terms of the order a/r and ignore higher powers of a/r. Calculate the potential due to this charge ring in space and obtain the field from the potential by taking the gradient *i.e.* by using the formula $\vec{E} = -\nabla\phi$.

9. Water flows through a pipe which undergoes a change in area from A to a (with $a < A$). Imagine a uniform flow away from the intersection region. Setup a divergence free velocity field in the entire pipe. You can assuming that the pipe is along the z-axis and the cross section of the pipe is in the $x - y$ plane.